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Subject – .NET Core

DATE- 29/07/2025

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Practical Assignment-01

1. Write a C# Program to perform basic arithmetic operations on two numbers.

Test Cases:

Inputs	Operation	Expected Output
10, 5	+ - * / %	Sum = 15, Diff = 5, Product = 50, Division = 2, Mod = 0
20, 6	+ - * / %	Sum = 26, Diff = 14, Product = 120, Division = 3, Mod=2

2. Write a C# Program to find average of three numbers.

Test Cases:

Inputs	Expected Output
10, 20, 30	Average = 20
5, 5, 10	Average = 6.66

3. Write a C# Program to calculate perimeter (circumference) of a circle.

Test Cases:

Radius	Expected Output
7	Circumference = 43.96
10	Circumference = 62.83

4. Write a C# Program to calculate area of a circle.

Test Cases:

Radius	Expected Output
7	Area = 153.94
10	Area = 314.16

5. Write a C# Program to calculate area of a triangle.

Test Cases:

Base	Height	Expected Output
10	5	Area = 25
6	8	Area = 24

6. Write a C# Program to calculate area of a square.

Test Cases:

Side	Expected Output
4	Area = 16
10	Area = 100

7. Write a C# Program to find max number between two numbers using ternary operator.

Test Cases:

A	B	Expected Output
5	10	Max = 10
20	15	Max = 20

8. Write a C# Program to check if a number is positive or negative, find max between two numbers, and check if entered year is a leap year.

Test Cases:

Input	Case	Expected Output
-5	Positive/Negative	Negative
10, 20	Max of Two	20
2024	Leap Year	Leap Year
2023		Not a Leap Year

9. Write a C# menu-driven program to perform basic arithmetic operations using switch statement.

Test Cases:

A	B	Choice	Expected Output
10	5	+	15
10	2	/	5
15	7	%	1

10. Write a C# Program to find the area of circle, square, and rectangle using switch statement.

Test Cases:

Shape	Input	Expected Output
Circle	Radius = 7	153.94
Square	Side = 4	16
Rectangle	L = 5, B = 6	30

11. Write a C# Program to generate electricity bill using else-if ladder.

Test Cases:

Units	Expected Output
80	₹400 ($80 \times ₹5$)
150	₹975 ($80 \times 5 + 70 \times 7$)
250	₹2000 (based on slabs)

12. Write a C# Program to calculate library fine amount using else-if ladder.

Test Cases:

Days Late	Expected Output
2	₹0
7	₹35 (5×7)
16	₹160 (10×16)

13. Write a C# Program to find minimum number between two numbers.

Test Cases:

A	B	Expected Output
10	5	5
8	8	Both Equal

14. Write a C# Program to find maximum number between two numbers.

Test Cases:

A	B	Expected Output
15	25	25
12	7	12

15. Write a C# Program to find minimum number between three numbers.

Test Cases:

A	B	C	Expected Output
4	8	6	4
12	9	15	9

16. Write a C# Program to find maximum number between three numbers.

Test Cases:

A	B	C	Expected Output
3	5	2	5
10	10	9	10

17. Write a C# Program to check if entered number is even or odd.

Test Cases:

Input	Expected Output
4	Even
11	Odd

18. Write a C# Program to check if number is even or odd using conditional operator.

Test Cases:

Input	Expected Output
6	Even
17	Odd

19. Write a C# Program to print odd numbers between 1 to 500.

Test Case:

Input	Expected Output
—	1, 3, 5,...499

20. Write a C# Program to print even numbers between 1 to 500.

Test Case:

Input	Expected Output
—	2, 4, 6,...500

21. Write a C# Program to add all odd numbers from 1 to 20.

Test Case:

Input	Expected Output
—	100

22. Write a C# Program to add all even numbers from 1 to 20.

Test Case:

Input	Expected Output
—	110

23. Write a C# Program to calculate grade of student based on obtained marks.

Test Cases:

Marks	Expected Output
85	Grade A
72	Grade B
60	Grade C
45	Grade D
30	Fail

24.WAP to demonstrate Type Inference.

25.WAP to demonstrate Console I/O operations in C#.

26.WAP to demonstrate printing concatenated string by using formatted string.

27.WAP to perform basic Arithmetic operations.

28.WAP to demonstrate Ternary Operator.

29.WAP to demonstrate Type Conversion.

30. Write a C# Program to find factorial of entered number.

Test Cases:

- **Input: 5**
Output: Factorial of 5 is ____
 - **Input: 0**
Output: Factorial of 0 is ____
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31. Write a C# Program to find sum of digits of entered number.

Test Cases:

- **Input: 123**
Output: Sum of digits is ____
 - **Input: 9876**
Output: Sum of digits is ____
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32. Write a C# Program to check if entered number is Armstrong or not.

Test Cases:

- **Input: 153**
Output: 153 is an Armstrong number.
 - **Input: 123**
Output: 123 is not an Armstrong number.
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33. Write a C# Program to find all Armstrong numbers between 1 to 500.

Test Cases:

- **Input: *(no input; static range from 1 to 500)***
Output:
 - **Armstrong numbers between 1 to 500:**
 - **____**
-

34. Write a C# Program to print reverse of entered number.

Test Cases:

- **Input: 1234**
Output: Reversed number is 4321_____
 - **Input: 100**
Output: Reversed number is 001 _____
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35. Write a C# Program to check if entered number is palindrome or not.

Test Cases:

- **Input: 121**
Output: 121 is a palindrome.
 - **Input: 123**
Output: 123 is not a palindrome.
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36. Write a C# Program to find all palindrome numbers between 1 to 500.

Test Cases:

- **Input: *(no input; static range from 1 to 500)***
Output:
 - **Palindrome numbers between 1 to 500:**
 - **_____**
-

37. Write a C# Program to display Fibonacci series up to given number.

Test Cases:

- **Input: 10**
Output: Fibonacci series up to 10: _____
 - **Input: 50**
Output: Fibonacci series up to 50: _____
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38. Write a C# Program to calculate gross salary; basic salary is entered by user.

Conditions:

- D.A = 40% of basic salary
- HRA = 20% of basic salary
- PPF = 10% of basic salary
- Gross Salary = Basic + DA + HRA - PPF

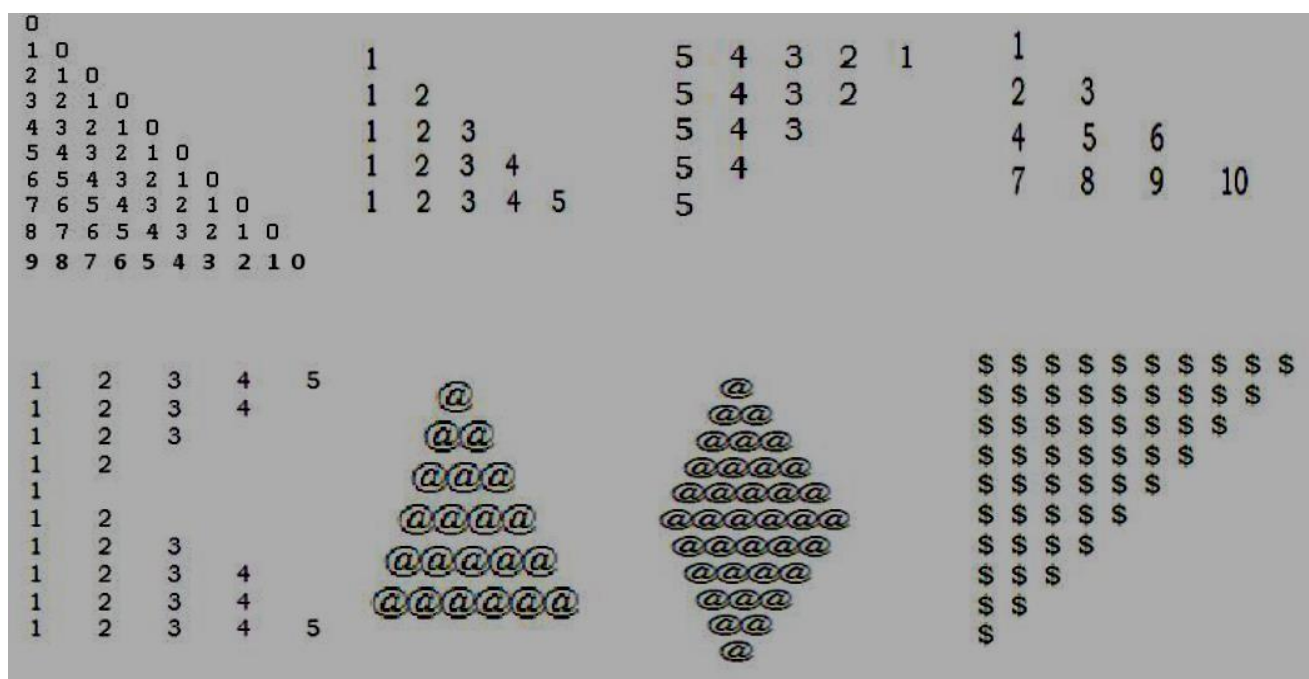
Test Cases:

- Input: 10000
Output: Gross Salary is 15000_____

39. Write a c# program to display Floyds Triangle.

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1
01
101
0101
10101
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40. Write a separate c# program to display each given pattern.



- 41. Write a C# Program to demonstrate Boxing and UnBoxing.**
- 42. Write a C# Program to demonstrate Enumeration.**
- 43. Write a C# Program to perform sum of array elements**
- 44. Write a C# Program to demonstrate various operations on Array using in built methods.**
- 45. Write a C# Program to demonstrates creation of Tuple using Parenthesis Method**
- 46. Write a C# Program to demonstrates creation of Tuple using Create() Method.**
- 47. Write a C# Program to demonstrate Nested Tuple.**
- 48. Write a C# Program to demonstrate Tuple as Method Parameter.**
- 49. Write a C# Program to demonstrate returning Tuple from Method.**
- 50. Write a C# program which stores even items & odd items in separate Arrays from 1D array**
- 51. Write a C# Program to perform Addition of TWO matrices.**